

Q1: Solution

The sequence of sets $\{C_k\}$, where $C_k = \{x : 1 - \frac{1}{k} \leq x \leq 2 + \frac{1}{k}\}$, is decreasing since for every $k = 1, 2, \dots$, $\frac{1}{k+1} < \frac{1}{k}$, and so, $1 - \frac{1}{k} < 1 - \frac{1}{k+1}$, and $2 + \frac{1}{k+1} < 2 + \frac{1}{k}$. Hence

$$\bigcup_{k=1}^{\infty} C_k = C_1 = \{x : 0 \leq x \leq 3\}$$

Q2: Solution

The sequence of sets $\{C_k\}$, where $C_k = \{z : 1 + \frac{1}{k} \leq z \leq 5 - \frac{1}{k}\}$, is increasing since for every $k = 1, 2, \dots$, $\frac{1}{k+1} < \frac{1}{k}$, and so, $1 + \frac{1}{k+1} < 1 + \frac{1}{k}$, and $5 - \frac{1}{k} < 5 - \frac{1}{k+1}$. Hence

$$\bigcup_{k=1}^{\infty} C_k = \lim_{k \rightarrow \infty} C_k = \{z : 1 < z < 5\}$$

Note that for every k , $z > 1$ and $z < 5$. Hence, the lower and upper bounds are not in the union.

Q3: Solution

If $A \subset B$, then $A \cap B = A$, and $A \cup B = B$. Hence $P(A \cap B) = P(A)$, and $P(A \cup B) = P(B)$. Also, for any arbitrary event C , $(A \cup C) \subset (B \cup C)$, and so $P(A \cup C) \leq P(B \cup C)$.

In addition, for any events A and B , since $A \cap B$ and $B \cap A^c$ are disjoint (mutually exclusive). Hence

$$P(B) = P((A \cap B) \cup (B \cap A^c)) = P(A \cap B) + P(B \cap A^c) = P(A) + P(B \cap A^c)$$

Q4: Solution

Let A and B be the events that Peter and Sarah drink coffee after their lunch, respectively. So, for example, if $P(A) = 0.4$, $P(B) = 0.45$, and $P(A \cap B) = 0.25$, we have

$$\begin{aligned} P(\text{At least one of them drink coffee}) &= P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ &= 0.4 + 0.45 - 0.25 = 0.6 \end{aligned}$$

$$P(\text{One, but not both, drinks coffee}) = P(A \cup B) - P(A \cap B) = 0.6 - 0.25 = 0.35$$

Q5: Solution

Let A_i , $i = 1, 2, 3, 4$, be the event that exactly i undergraduate student(s) is (are) in the group. Then

$$P(A_1) = \frac{\binom{4}{1}\binom{3}{3}}{\binom{7}{4}} \approx 0.114, \quad P(A_2) = \frac{\binom{4}{2}\binom{2}{2}}{\binom{7}{4}} \approx 0.514, \quad P(A_3) = \frac{\binom{4}{3}\binom{1}{1}}{\binom{7}{4}} \approx 0.343, \quad P(A_4) = \frac{\binom{4}{4}\binom{0}{0}}{\binom{7}{4}} \approx 0.029.$$